UE26MA141A Engineering Mathematics – I · Unit 1 · Partial Differentiation

Formulas and symbols

Every formula in scope, then a glossary of every symbol used.

Class 005 · Homogeneous functions and Euler's theorem · compiled 14 October 2026

ESA ★★★	Asked repeatedly in past ESA papers
ESA ★★	Asked in past ESA papers or computed in class-work problems
ESA ★	Conceptual: taught in class, not examined so far
ADDED FOR CLARITY	Standard context that is not in the class material
TRANSCRIPT ONLY	Said in the lecture but not on any slide
VIDEO ONLY	Seen only in a frame of the class video
NOT ON CURRENT SLIDES	From older material or past papers, not this year's slides
S005·6	Deck 005, slide 6 (rendered .pptx)
T005 00:08:57	AV-summary transcript 005, segment start
CW U1 P20	Unit 1 class-work problem 20
ESA Jan-24 s3 1c	Past paper: Jan 2024, set 3, question 1.c

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A Formula sheet

1 Homogeneous functions

E1.1 · Homogeneous of degree n

ESA ★★

$$u = x^n g\left(\frac{y}{x}\right) = y^n h\left(\frac{x}{y}\right) \quad \Longleftrightarrow \quad u(tx, ty) = t^n\, u(x, y)$$

GIVES	The definition and the equivalent scaling test.
APPLIES	Two variables; with three, use $x^n g(y/x, z/x)$.
NOTE	The scaling test is added for clarity; the slides use the $x^n g(y/x)$ form.

S005·3 · S005·4 · S005·5

E1.2 · Degree of a quotient and of a product ESA ★★

$$\deg \frac{P}{Q} = \deg P - \deg Q, \qquad \deg(PQ) = \deg P + \deg Q$$

GIVES Degree of combinations of homogeneous functions.

APPLIES P, Q homogeneous.

NOTE $\frac{\sqrt{x}+\sqrt{y}}{x+y}$: $\frac{1}{2} - 1 = -\frac{1}{2}$. A ratio such as $\sin^{-1}(y/x)$ has degree 0.

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2 Euler's theorem

E2.1 · Euler's theorem, two variables ESA ★★★

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n\,u$$

GIVES The scaled sum of first partials.

APPLIES u homogeneous of degree n.

NOTE Compact form: $xu_x + yu_y = nu$.

S005·6 · T005 00:08:57

E2.2 · Euler's theorem, three variables ESA ★★

$$x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} + z\frac{\partial u}{\partial z} = n\,u$$

GIVES The same result with one term per variable.

APPLIES $u = f(x, y, z)$ homogeneous of degree n.

NOTE Extends to any number of variables.

S005·6 · T005 00:14:28

3 Second and higher order: the corollary

E3.1 · Corollary: second order ESA ★★★

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\,\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = n(n-1)\,u$$

GIVES The scaled sum of second partials.

APPLIES u homogeneous of degree n with continuous second partials.

NOTE Coefficients $1, 2, 1$ as in $(x+y)^2$.

S005·6 · T005 00:12:21

E3.2 · Third order

ESA ★

$$x^3 u_{xxx} + 3x^2 y\, u_{xxy} + 3xy^2 u_{xyy} + y^3 u_{yyy} = n(n-1)(n-2)\, u$$

GIVES The pattern continues with binomial coefficients.

APPLIES Third partials continuous.

NOTE The lecture states the right-hand side; the coefficients 1, 3, 3, 1 are added for clarity.

TRANSCRIPT ONLY ADDED FOR CLARITY T005 00:12:43 · T005 00:14:12

E3.3 · First plus second order

ESA ★★★

$$x^2 u_{xx} + 2xy\, u_{xy} + y^2 u_{yy} + x\, u_x + y\, u_y = n(n-1)u + nu = n^2 u$$

GIVES The combination asked in ESA Jan 2024 sets 3 and 4.

APPLIES u homogeneous of degree n; for $u = u_1 + u_2$ apply to each part.

NOTE Degrees 3 and -3 both give 9: so $9u$ for $x^3 \sin^{-1}\frac{y}{x} + y^{-3}\tan^{-1}\frac{x}{y}$.

ADDED FOR CLARITY ESA Jan-24 s3 1c · ESA Jan-24 s4 1d · CW U1 P23

4 Extension: when f(u) is homogeneous

E4.1 · Extension, first order

ESA ★★★

$$f(u) \text{ homogeneous of degree } n \implies x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = n\frac{f(u)}{f'(u)} = g(u)$$

GIVES Euler's combination for a function that is not itself homogeneous.

APPLIES Some $f(u)$ is homogeneous (e.g. $\tan u$, $\sin u$, e^u).

NOTE $\tan u$: $g = n\tan u\cos^2 u = \frac{n}{2}\sin 2u$.

S005·7

E4.2 · Extension, second order

ESA ★★★

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x\,\partial y} + y^2\frac{\partial^2 u}{\partial y^2} = g(u)\left[g'(u) - 1\right]$$

GIVES The second-order combination for the extension.

APPLIES $g(u) = n\,f(u)/f'(u)$ from E4.1.

NOTE Differentiate g with respect to u.

S005·7

B Symbol glossary

Symbol	Name	Meaning	Appears in
$u = f(x, y)$	Function	The function tested for homogeneity	E1.1, E2.1
n	Degree	Power in $x^n g(y/x)$; any real number	E1.1–E4.2
$g(\cdot)$ (topic 1)	Arbitrary function of ratios	$u = x^n g(y/x)$	E1.1
$g(u)$ (topic 4)	Extension function	$nf(u)/f'(u)$ (a different g from topic 1)	E4.1, E4.2
$f(u)$	Homogeneous function of u	e.g. $\tan u$ when $u = \tan^{-1}(\dots)$	E4.1
t	Scale factor	$u(tx, ty) = t^n u$	E1.1
u_x, u_y, u_z	First partials	Homogeneous of degree $n - 1$	E2.1, E2.2
u_{xx}, u_{xy}, u_{yy}	Second partials	Degree $n - 2$; $u_{xy} = u_{yx}$	E3.1
u_1, u_2	Homogeneous parts	Parts of different degrees in a sum	E3.3
P, Q	Homogeneous factors	Numerator and denominator of a quotient	E1.2

Remark. The letter g has two meanings: the arbitrary function of ratios in $x^n g(y/x)$ (topic 1), and $g(u) = nf(u)/f'(u)$ in the extension (topic 4).